

Depth, Breadth, and Bias: Structural Diffusion of Political Content on Divergent Platforms

Anonymous submission

Abstract

When political content spreads on social media, does diffusion follow universal structural patterns or are these patterns shaped by platform-specific communities? We address this question by comparing how cascades during the 2024 U.S. presidential campaign scale on two ideologically distinct but technically comparable platforms: Truth Social and Bluesky. Using repost and reply networks constructed from public posts mentioning Biden or Trump (May–June 2024), we examine whether similarly sized cascades take similar shapes across platforms. We find that amplification (reposts) scales similarly. Conversational diffusion, however, diverges sharply: reply cascades on Truth Social are wider and shallower, while those on Bluesky are narrower and deeper. We evaluate candidate explanations: influencer dominance, topic variability, ideological composition, and cross-ideological interaction alignment. Neither influencers nor topic differences account for this divergence. Only when both ideological composition and interaction alignment are modeled together [do these platform-level scaling differences largely attenuate](#). Fine-grained motif analysis further reveals asymmetric interaction patterns, including distinct roles for left-, right-, and center-leaning users across platforms. [Overall, the results suggest that political information can be amplified similarly across platforms even when conversational engagement is organized differently across ideological groups.](#)

1 Introduction

Social media platforms facilitate information cascades, reposts, replies, and other interactions through which political content circulates online (Easley and Kleinberg 2010; Gruhl et al. 2004; Guille et al. 2013). A central question in computational social science is whether the structural patterns of these cascades are governed by universal dynamics or shaped by the particular environments in which they unfold. While there is past work on how diffusion structures vary across information types (e.g., false vs. real news (Vosoughi, Roy, and Aral 2018)), we know little about how diffusion scaling properties vary across platforms.

This question has become increasingly relevant because the social media ecosystem has fractured. After the January 6 Capitol attack, Donald Trump launched Truth Social as a right-leaning alternative to mainstream platforms. More recently, Twitter’s rebranding as X and its loosening of moderation have spurred many centrist and left-leaning users to

migrate to Bluesky (Gerard, Botzer, and Weneringer 2023; Failla and Rossetti 2024; Quelle and Bovet 2025). Although both platforms mimic Twitter’s design, follower networks, reposts, and reply mechanics, their contrasting community compositions create distinct social contexts. This creates a natural testbed to determine whether similar technical designs produce similar diffusion structures when embedded in different communities.

To answer this question, we compare the structural diffusion of political content on Truth Social and Bluesky during a high-salience period of the 2024 U.S. presidential campaign. The analysis centers on a one-month window that included major events such as the first presidential debate, focusing on posts containing hashtags and keywords related to “Biden” and “Trump.” To characterize diffusion dynamics, we go beyond raw comparisons of cascade size, depth, or breadth. Such comparisons across platforms can be misleading since platforms with more users mechanically generate larger cascades (Juul and Ugander 2021). Instead we focus on how cascades grow as they scale. [We compute scaling to determine whether cascades tend to broaden horizontally or deepen vertically as they grow in the number of posts.](#) If scaling relationships are platform-specific, identical levels of engagement produce fundamentally different conversational geometries. Our goal is to understand if such divergence happens and why.

When performing cross-platform comparisons, we also distinguish between diffusion modes. Reposts and replies serve different functions: reposts are acts of amplification that redistribute content to new audiences, while replies are conversational acts that extend, contest, or elaborate on content (Conover et al. 2011; boyd, Golder, and Lotan 2010; Guerrero-Solé 2018). Past work has identified important distinctions between these diffusion modes, such as differences in homophily (Conover et al. 2011). However, cascade research typically aggregates both into general engagement metrics. By treating repost and reply cascades as distinct objects, we can test whether cross-platform universality holds for amplification, for conversation, or for both.

We find that platform differences matter most where political diffusion becomes conversational. Repost cascades scale similarly across Truth Social and Bluesky. Reply cascades, by contrast, diverge sharply: Truth Social produces wider and shallower interactions, while Bluesky produces

narrower and deeper conversational threads.

What explains these fundamental differences? Platforms can vary in the dominance of their influential users and this can impact diffusion patterns (Goel et al. 2016; Zannettou et al. 2018). Similarly, certain topics can be more contentious, leading to different diffusion patterns (Vosoughi, Roy, and Aral 2018; Wang et al. 2022) and topic distribution can vary across platforms. Interestingly, we find neither of these explain the reply scaling differences between the two platforms. Ideological composition partially explains the differences and, most importantly, when accounting for both composition and the degree to which users engage in cross-ideological discussions, [most of the platform-level difference is accounted for](#).

Motif analysis helps unpack what this aggregate alignment pattern means locally. It shows that cross-ideological interaction takes different structural forms across platforms: on Truth Social, disagreement more often appears as parallel replies around the same parent post, whereas on Bluesky it more often appears as sequential chains that extend conversations across turns.

These findings matter for understanding the fragmented social media ecosystem. Newer platforms are often built around familiar Twitter-like features, but similar technical designs do not guarantee similar political interaction. Our findings show that political information can be amplified in similar ways while being discussed in very different ways, with replies revealing how platform communities organize opportunities for ideological engagement, disagreement, and visibility.

2 Related Work

2.1 Comparative Cross-Platform Analysis

A growing body of research argues that social media should be studied as a fragmented ecosystem of platforms rather than as a single, unified public sphere (Rowe and Alani 2014; Boulianne and Larsson 2023). Comparative work shows that political ideology predicts both platform adoption and political use, while platform-specific networking features shape what kinds of political exposure and expression occur there (Boulianne, Hoffmann, and Bossetta 2025). At the ecosystem level, recent scholarship suggests that online fragmentation increasingly takes the form of “echo platforms,” in which ideological differentiation emerges not only within platforms but between them as users migrate across services with distinct moderation regimes, recommendation logics, and community norms (Di Martino et al. 2025; Mosleh, Allen, and Rand 2025).

This perspective is especially important for the study of alternative and emerging platforms. Research on Gab established early that alt-tech platforms can attract ideologically concentrated user bases and sustain different information environments from those observed on mainstream sites (Zannettou et al. 2018). More broadly, Cinelli et al. (2021) show that homophilic clustering and like-minded information diffusion vary substantially across platforms, indicating that similar affordances do not guarantee similar interaction dynamics. Their comparison of Facebook, Twitter, Reddit,

and Gab underscores the importance of platform-specific interaction environments in shaping information exposure and segregation.

Work on Truth Social remains comparatively sparse, but existing studies already suggest that it is best understood as a politically concentrated alt-tech environment rather than simply a new campaign tool. Early dataset work positioned Truth Social as a refuge for users alienated by mainstream moderation and documented its rapidly emerging network and content ecology (Gerard, Botzer, and Weninger 2023). Subsequent work released a large 2024 election dataset with posts, replies, and user interactions, emphasizing the platform’s role in hosting contentious, often hyper-partisan political discourse (Shah et al. 2024). More recent network analysis similarly characterizes Truth Social as ideologically homogeneous and structurally segmented, with attention concentrated around a small set of central actors (Hughes and Weninger 2025).

By contrast, recent work on Bluesky describes a Twitter-like platform with a markedly different political center of gravity. Early studies characterized Bluesky as an emerging alternative to X and documented its rapidly expanding data ecology, including large-scale post histories, follow relations, replies, reposts, quotes, and feed-generator interactions (Failla and Rossetti 2024). More recent work shows that Bluesky exhibits heavy-tailed, highly clustered network structure similar to larger platforms, but that users share predominantly left-center news sources and make limited use of custom algorithmic feeds (Quelle and Bovet 2025). Nogra et al. (2026) further find that Bluesky maintains relatively high levels of original posting, low toxicity, and high-credibility news sharing even after rapid growth following its public launch. Together, these studies position Truth Social and Bluesky as a valuable comparative pair: technically comparable platforms embedded in sharply different ideological communities.

2.2 Information Diffusion and Cascade Structure

Research on information diffusion has developed a rich vocabulary for describing how online content spreads through networks, including cascade size, depth, breadth, and structural virality (Gruhl et al. 2004; Guille et al. 2013; Easley and Kleinberg 2010; Goel et al. 2016). This literature has linked cascade structure to content attributes such as veracity, novelty, emotional tone, and topic salience. Most prominently, Vosoughi, Roy, and Aral (2018) show that false news on Twitter spreads farther, faster, deeper, and more broadly than true news. Further work examines how topic salience (Wang et al. 2022), cascade predictability (Cheng et al. 2014), and misinformation propagation (Zhao et al. 2020) shape diffusion. [Recent work continues to separate these dimensions, distinguishing broadcast-like spread from long peer-to-peer chains \(González-Bailón et al. 2024\) and depth from breadth when assessing platform interventions \(Slaughter et al. 2025\). A parallel line asks whether such structures can be measured when platforms do not record diffusion paths: DeVerna et al. \(2024\) show for Twitter and Bluesky that reconstruction assumptions can distort the resulting cascades.](#)

A related development concerns interaction type. Diffusion studies often prioritize reposting or retweeting because these traces are easy to reconstruct and map directly onto dissemination. Reposting is especially important for political amplification: experimental evidence from Facebook shows that reshared content substantially increases exposure to political news and low-quality sources, demonstrating the centrality of reshare mechanisms to political visibility (Guess et al. 2023a). Foundational work on retweeting also shows that reposts are socially meaningful acts of redistribution, attribution, and audience management, rather than neutral transmission alone (boyd, Golder, and Lotan 2010; Park and Kaye 2019). [Who does the resharing also matters: on Twitter/X, a small elite of high-influence accounts carries roughly half of all repost-cascade flow \(Niitsuma et al. 2025\).](#)

Yet reposting is only one mode of information flow. Prior work shows that retweets, mentions, replies, and quote tweets capture different forms of political engagement. In U.S. election discourse, retweet networks are highly segregated by ideology, whereas mention networks are more politically heterogeneous and contain more cross-ideological interaction (Conover et al. 2011). Studies of political discussions similarly treat retweeting, mentioning, and replying as distinct behavioral traces, with retweet networks more likely to resemble echo chambers and replies or mentions more likely to reflect direct interaction (Guerrero-Solé 2018). Other work finds that emotional and textual features affect replies and retweets differently, suggesting that response-oriented conversation and amplification-oriented redistribution follow different behavioral logics (Kim and Yoo 2012; Toraman et al. 2022). Finally, studies of threaded political conversations show that replies can sustain extended argument across ideological divides, making reply trees especially useful for studying conversational engagement rather than simple visibility (Shugars and Beauchamp 2019).

2.3 Ideology, Polarization, and Political Communication Online

A large literature shows that online political communication is shaped by ideological homophily, selective exposure, and asymmetric cross-group engagement (Adamic and Glance 2005; Conover et al. 2011; Flaxman, Goel, and Rao 2016; Del Vicario et al. 2016). Yet this work also suggests that different interaction types produce different political structures. Conover et al. (2011) show that retweet networks are highly segregated, whereas mention networks are more politically heterogeneous. Similarly, Barberá et al. (2015) find that political communication often remains ideologically clustered, but that liberals are more likely than conservatives to engage in cross-ideological dissemination. These studies suggest that exposure to disagreement is unevenly distributed and depends in part on how users interact.

More recent work complicates any simple equation of cross-cutting exposure with deliberative benefit. Bail et al. (2018) show that following opposing political elites on social media can increase polarization rather than reduce it. Garimella et al. (2018) further identify gatekeepers and the “price of bipartisanship,” showing that users who bridge po-

larized communities often occupy structurally costly positions. In parallel, Rathje, Van Bavel, and van der Linden (2021) demonstrate that out-group animosity is a strong driver of sharing, implying that cross-ideological interaction on social media is often conflictual rather than integrative.

Recent evidence also points to ideological asymmetries in what gets diffused. Chang et al. (2025) show that liberals and conservatives differ in the range of issues and the toxicity of elite content they retransmit, suggesting that downstream information environments are shaped not only by whom users follow, but also by what kinds of content each side disproportionately passes along. This raises a related question for cascade analysis: whether political structure operates mainly through user composition, through the alignment of interactions within and across groups, or through both at once.

2.4 Theoretical Framework and Hypotheses

Defining Diffusion Modes and Scaling. As we mentioned before, we distinguish between two modes of diffusion: reposts and replies. Rather than emphasizing only how far or how fast information travels, our analysis focuses on the shapes of cascades: whether content spreads broadly through direct amplification or develops sequentially through extended exchanges. Because these structural measures all correlate with cascade size (Juul and Ugander 2021), we hold size constant throughout and examine how cascades grow vertically (depth) and horizontally (breadth).

This scaling perspective leads to a broader question: when cascades reach similar sizes, do they take similar structural forms across platforms and diffusion modes, or do they grow through different geometric patterns? If diffusion scaling is broadly universal, similarly sized cascades should exhibit comparable depth and breadth regardless of platform. If scaling is platform-specific, then the same level of participation may produce different cascade shapes depending on the interaction environment.

We therefore examine three candidate explanations for cross-platform variation in cascade geometry. First, user prominence may shape scaling if highly followed or highly active users attract many direct interactions, producing broader cascades. Second, topic variability may matter if some topics invite rapid amplification while others encourage extended discussion. Third, ideological composition and interaction alignment may shape scaling by affecting whether interactions remain within like-minded groups or extend across ideological boundaries.

H1: Influencer Dominance. Prior research has demonstrated that a small number of high-follower users can disproportionately influence content dissemination, particularly on platforms characterized by skewed follower distributions (Goel et al. 2016; Zannettou et al. 2018). Truth Social, with its relatively small and ideologically concentrated user base, offers a fertile environment for such dynamics. This is supported by power-law distributions in followers, reposts, and replies per user (see Appendix A), where Truth Social exhibits a more pronounced heavy-tailed distribution than Bluesky, with a handful of dominant accounts occupying central roles. [A clear discontinuity separates the top](#)

eight Truth Social accounts from the rest (the largest has 56 times the followers of the platform’s 99.9th-percentile account), while Bluesky shows no such break, so we treat these eight as influencers instead of removing the top k accounts on both platforms (Appendix A). We hypothesize that this concentration contributes to reply cascades that are broad but shallow, indicative of immediate visibility rather than sustained conversational engagement.

H2: Ideological Structure. The ideology of users in a cascade may shape its structure through both participation and interaction. Prior work shows that online political interaction is strongly structured by ideological homophily and selective exposure: amplification networks are often more segregated, while more conversational channels can include greater cross-ideological contact (Conover et al. 2011; Garimella et al. 2018; Barberá et al. 2015; Adamic and Glance 2005; Guess et al. 2023b). However, this literature does not directly establish how ideological alignment affects cascade geometry. We build on it by asking whether ideological composition and cross-ideological interaction are associated with how reply cascades scale in depth and breadth.

We measure ideological structure along two dimensions: (1) *ideological composition*, operationalized as the proportion of posts in a reply tree authored by users in each ideological category, and (2) *interaction alignment*, defined as the share of reply edges connecting users with the same ideological label. Lower alignment indicates more cross-cutting interaction. As shown later in Appendix B, Truth Social and Bluesky differ substantially in the ideological composition of their observed users, making ideological structure a plausible source of cross-platform differences in reply geometry.

We hypothesize that ideological composition and interaction alignment play a significant role in how reply cascades scale. If cross-platform differences in cascade geometry are driven by ideological structure, then incorporating composition and alignment into the models should substantially attenuate platform-level differences in depth and breadth.

H3: Topic Variability. Post content shapes how users interact with it. Emotionally charged, controversial, or conspiratorial content often spreads through deeper and more structurally viral cascades than neutral or factual information (Vosoughi, Roy, and Aral 2018; Wang et al. 2022). As described in Section 3.4, we identified eleven salient topics across both platforms; while all are politically related, their distribution differs substantially between Truth Social and Bluesky. We hypothesize that topics play a significant role in how diffusion dynamics scale on social media. For instance, topics involving partisan conflict, institutional distrust, or election-related narratives may elicit distinct structural diffusion patterns depending on their resonance within each platform’s community.

Micro-Level Network Motifs. To unpack the mechanisms behind platform-level differences in reply cascade geometry, we complement cascade-level regressions with motif analysis, a graph-theoretic approach examining the frequency of small recurring substructures within larger networks (Alon 2007). While statistical models evaluate

whether alignment ratio and ideological composition explain divergence, they do not reveal *how* these compositional factors shape micro-dynamics of interaction. We analyze two basic 3-node motif shapes: *chains*, where replies unfold in sequence, and *stars*, where multiple replies attach to the same root. Each motif is labeled by the ideological stance of its participants: Left (L), Center (C), or Right (R).

3 Data and Methods

3.1 Data Collection

We collected posts and their associated reposts and replies from both platforms during a critical month of the 2024 U.S. presidential campaign, when major political events (including the first debate) spurred intense online discussion. We used each platform’s public API to collect posts containing the target keywords and their surrounding conversations. For Bluesky, the thread endpoint enabled retrieval of entire discussion trees around posts mentioning “Biden” or “Trump,” while for Truth Social (which follows the Mastodon API structure), we used the context endpoint to capture both ancestor and descendant replies recursively. This approach provides full conversational context rather than isolated messages, allowing us to reconstruct entire reply cascades. For every post and reply, we also collected associated repost actions and each user’s follower list. Data were collected between May 30 and June 30, 2024, a high-salience period that included the first U.S. presidential debate.

3.2 Cascade Construction

We model information diffusion as directed acyclic graphs (DAGs), where each node is a single post (the root post, a reply, or a repost) and edges represent inferred diffusion pathways, following the time-inferred approach of Vosoughi, Roy, and Aral (2018). Each node is thus one action by a user, and a user who replies several times in one thread contributes one node per reply. We keep these appearances separate because a reply tree records how a conversation actually unfolded; merging them would add a modeling assumption to observed data. Ideology is aggregated at the user level: each node carries the label (left-leaning, center, or right-leaning) of the account that wrote it (see Ideology Estimation and Alignment), so a cascade’s ideological composition is the share of its posts written by accounts in each category. For **reply cascades**, trees were constructed directly from reply metadata: each reply is linked to its parent post, and cascades are rooted at the original post with subsequent replies linked recursively. For **repost cascades**, platform APIs report all reposts as nominally linking to the original post, which obscures the true diffusion path. To approximate true repost trees, we linked each repost r_i to its earliest possible source, as in the “first influence” rule of Bakshy et al. (2011): the original post if the user who made r_i is observed to follow its author in the follower snapshot, and otherwise the earliest prior repost r_j with $t_j < t_i$ made by a user they are observed to follow, giving the edge $r_j \rightarrow r_i$. When no such user existed, the repost was attributed directly to the original post. Because these edges are inferred and reconstruction

assumptions can distort cascade properties (DeVerna et al. 2024), we refer to repost cascades as reconstructed through-out and, as in that earlier work, verify that the results hold under alternative attribution rules (Appendix, Robustness of Repost Cascade Reconstruction). The follower network is a snapshot without edge creation dates, collected together with the posts and reposts, one day apart, and at the same time for both platforms.

3.3 Cascade Metrics

For each cascade we compute four structural metrics. **Size** (N) is the total number of posts in the cascade, including the root. **Breadth** (B) is the maximum number of nodes at any single depth level: $B = \max_{d \geq 1} n(d)$, where $n(d)$ is the node count at depth d . This captures the widest “layer” of the cascade and differs from root out-degree, which counts only direct children of the root. **Depth** (D) is the maximum hop distance from the root to any leaf node: $D = \max_v \text{dist}(\text{root}, v)$. **Structural Virality** (Goel et al. 2016) is the average pairwise shortest-path distance between all nodes in the cascade, distinguishing broadcast-like (low virality) from chain-like viral (high virality) spread. All metrics were log-transformed for modeling due to heavy-tailed distributions.

3.4 Topic Modeling

To identify dominant themes and evaluate H3, we applied BERTopic, which integrates contextual sentence embeddings (all-mpnet-base-v2), UMAP dimensional-ity reduction, and HDBSCAN clustering. Topics were extracted using class-based TF-IDF over a preprocessed vocabulary. Two human annotators independently reviewed a sample of representative posts from each cluster to produce consensus-based labels, yielding eleven coherent topic categories (inter-annotator $\kappa = 0.53$; model–Human 1 $\kappa = 0.67$; model–Human 2 $\kappa = 0.67$). Posts not assigned to any cluster were excluded from topic analyses. While both platforms engage heavily with Trump’s legal cases and the 2024 election, Bluesky posts are more concentrated around policy and legal accountability, whereas Truth Social emphasizes celebratory or pro-Trump content (see Figure 1).

3.5 Ideology Estimation and Alignment

Post-Level Stance Detection. We used the Llama-3.1-8B-Instruct model to classify each reply into one of five ideological stances (*left*, *lean-left*, *center*, *lean-right*, *right*), providing the original post and preceding replies as context. For robustness, we grouped stances into three categories: *left-leaning* (*left*, *lean-left*), *center*, and *right-leaning* (*lean-right*, *right*). Annotations were validated against two human annotators on a random sample of 200 replies. Agreement between the model and each annotator ($\kappa = 0.61$ and 0.74) is largely comparable to agreement between the two annotators ($\kappa = 0.77$), and the robustness checks in Appendix B show that our results hold under alternative ideology thresholds and under label errors at the measured rates. Because large language models may encode biases, we treat these labels as probabilistic indicators rather than ground truth.

User-Level Ideology Inference. User ideology was inferred by aggregating stance labels across all posts authored by each user. A user was classified as *left-leaning* if more than 60% of their posts were labeled left or lean-left, *right-leaning* if more than 60% were labeled right or lean-right, and *center* otherwise. Center-labeled users are therefore those whose posts alternate between left and right, together with those who consistently post without an overt ideological stance. This threshold balances ideological signal with label consistency; aggregate patterns are robust across thresholds from 0.5 to 0.7 (see Appendix B).

Alignment Ratio. For each reply edge, we code it as *aligned* if both users share the same ideology and *misaligned* otherwise. The alignment ratio for a cascade is the proportion of aligned reply edges. High ratios indicate ideologically homogeneous discussions; lower ratios reflect cross-cutting exchanges. In this scenario, consider a thread whose root post is by a right-leaning user A . Two right-leaning users, B and C , reply to A ; a left-leaning user D also replies to A ; and another left-leaning user E replies to D . The replies from B , C , and E are aligned and the reply from D is misaligned, so the alignment ratio is $3/4 = 0.75$. The measure describes the conversation: the same user can contribute aligned edges in one thread and misaligned edges in another.

3.6 Statistical Modeling

Baseline Model. To rigorously assess how cascade structure varies across platforms, we model the scaling relationship between cascade size and breadth using robust regression with the Huber loss function. This approach mitigates the influence of outliers, common in heavy-tailed cascade data (see Appendix H for diagnostics). Robust estimation bounds the influence of extreme residuals without correcting non-constant error variance, so standard errors come from the sandwich covariance estimator; we use Huber’s proposal 2 scale, since the default degenerates here.

Our baseline model examines how cascade breadth and depth scale with size, conditional on platform, including a platform indicator and its interaction with log-transformed cascade size:

$$\begin{aligned} \log(Y_i) = & \beta_0 + \beta_1 \log(\text{size}_i) \\ & + \beta_2 \cdot \text{platform}_i \\ & + \beta_3 \cdot (\log(\text{size}_i) \times \text{platform}_i) + \varepsilon_i, \end{aligned} \quad (1)$$

where Y_i denotes either cascade breadth or depth. The coefficient of interest is β_3 , which captures whether the relationship between cascade size and breadth differs by platform.

To quantify the explanatory contribution of additional covariates, we track percent attenuation of β_3 :

$$\Delta = 1 - \frac{\hat{\beta}_{3,\text{aug}}}{\hat{\beta}_{3,\text{base}}}, \quad (2)$$

where $\hat{\beta}_{3,\text{base}}$ is the interaction coefficient from Eq. 1 and $\hat{\beta}_{3,\text{aug}}$ is the corresponding estimate from augmented speci-

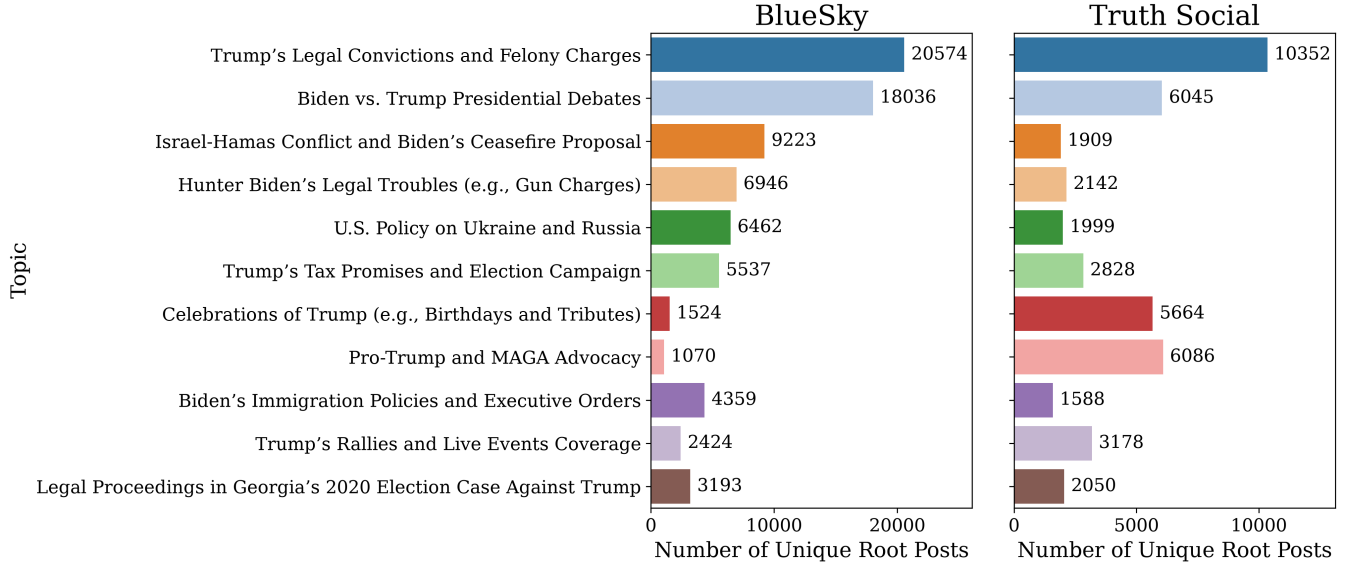


Figure 1: Topic distribution across *Bluesky* and *Truth Social*. Bar plots show the number of unique root posts assigned to each of the 11 topics, based on BERTopic clustering and manual labeling. The panels use independent x -scales (Bluesky $n = 79,348$, Truth Social $n = 43,841$); compare across panels using the printed counts.

fications (Models 2–4). A larger Δ indicates that added covariates account for a greater portion of cross-platform difference.

Incorporating Influencer Dynamics. To evaluate H1, we extend the platform variable to three categories: 0 for Bluesky, 1 for Truth Social cascades initiated by non-influencers, and 2 for those initiated by the top eight high-follower users on Truth Social. No comparable concentration of outlier accounts exists on Bluesky.

Assessing Ideological Composition and Alignment Effects. Building on user-level ideology labels, we include alignment ratio (spline-transformed, $df=3$, to capture non-linear effects at extremes of homogeneity) and the proportions of left- and right-leaning users per reply tree (with a spline for the left-user ratio and a linear term for the right-user ratio, based on partial-residual diagnostics; see Appendix I). Both measures are interacted with cascade size and platform indicators. We decompose this into three specifications: composition only (Model 3a), alignment only (Model 3b), and combined composition and alignment (Model 3c).

Evaluating Topic Variability. To address H3, we include cascade-level topic annotations as categorical moderators interacted with $\log(\text{size}_i)$ and platform_i , testing whether content differences account for cross-platform variation in how depth and breadth scale with size.

3.7 Network Motif Analysis

As introduced above, we use motif analysis to move from cascade-level regression results to the local interaction patterns that compose reply cascade geometry. Here, we describe how motifs are defined, counted, randomized, and

scored. We focus on directed 3-node tree motifs, which capture the simplest local distinction between branching and sequential interaction. We distinguish between two topological forms: **star motifs**, where one parent node has two child nodes, and **chain motifs**, where three nodes form a directed path of length two. Each node is annotated with the corresponding user’s ideology label: Left, Center, or Right. Motifs are treated as ordered configurations. For chain motifs, order follows the direction of reply; for star motifs, the parent position is distinguished from the child positions. We also summarize ordered motifs into broader families: *aligned* motifs, in which all three users share the same ideology; *one-different* motifs, in which two users share one ideology and the third differs; and *all-different* motifs, in which all three ideological groups appear.

Motifs are counted within each reply tree and then aggregated by platform. For each observed reply tree, the null model preserves the number of nodes and edges while randomizing edge assignments among users within the tree; node ideology labels are held fixed. This preserves the number of cascades, cascade sizes, and platform-specific ideological composition while randomizing local attachment patterns. For each platform, we generate $R = 100$ randomized versions of every cascade by swapping which posts replies attach to, keeping each post’s number of replies and every user’s ideology fixed (Appendix, Motif Frequencies and the Null Model). For motif type i , we compute the standardized motif score:

$$Z_i = \frac{N_{\text{obs}}(i) - \mu_{\text{null}}(i)}{\sigma_{\text{null}}(i)},$$

where $N_{\text{obs}}(i)$ is the observed count of motif i , and $\mu_{\text{null}}(i)$ and $\sigma_{\text{null}}(i)$ are the mean and standard deviation of its

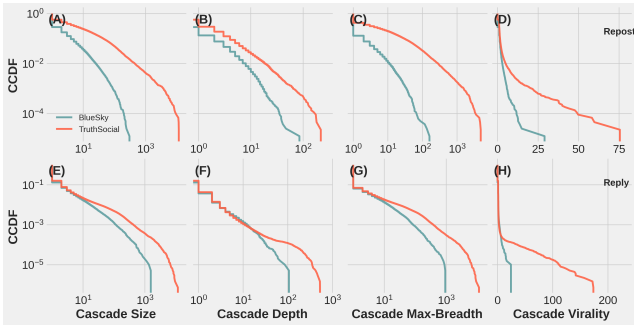


Figure 2: (A–D) Complementary cumulative distribution functions (CCDFs) of repost cascade statistics across two platforms: BlueSky and TruthSocial. Each subplot represents a different structural property: Size, Depth, Breadth, and Structural Virality (Goel et al. 2016). (E–H) The same analyses applied to reply cascades using the same metrics. X-axes are on a log scale (except for structural virality), and y-axes are on a log scale throughout. Each line represents a platform’s empirical distribution.

count across randomized realizations. Positive values indicate overrepresentation relative to the null, while negative values indicate underrepresentation. We interpret motif scores by their relative direction and magnitude across platforms rather than treating the motif panel as a set of independent confirmatory hypothesis tests.

4 Results

4.1 Diffusion Structure Across Emerging Social Networks

As shown in Figure 2, cascades on Truth Social appear larger, deeper, and broader than those on Bluesky for both reposts and replies, and show higher levels of structural virality. A straightforward explanation is platform size, since larger user bases mechanically allow cascades to expand further. Prior work has shown that depth, breadth, and structural virality all correlate positively with cascade size (Juul and Ugander 2021), which raises a concern that marginal comparisons conflate structural effects with sheer volume. To address this, we hold cascade size constant and examine structural differences directly.

Table 1 gives the counts behind these distributions. Most Bluesky reply cascades (71.3%) are a single post, against 43.3% on Truth Social, and 9.0% of Truth Social reply cascades exceed 50 posts, against 0.3% on Bluesky; repost cascades are more alike in size.

Figure 3 presents robust linear fits between log-transformed cascade size and both depth and breadth. After conditioning on cascade size, differences in repost behavior are small: repost cascades scale similarly across Truth Social and Bluesky, with nearly identical slopes. In contrast, reply cascades exhibit a marked divergence. On Truth Social, replies tend to fan out, producing broad but shallow trees, whereas on Bluesky, conversations deepen through sequential exchange, forming narrower but more layered structures.

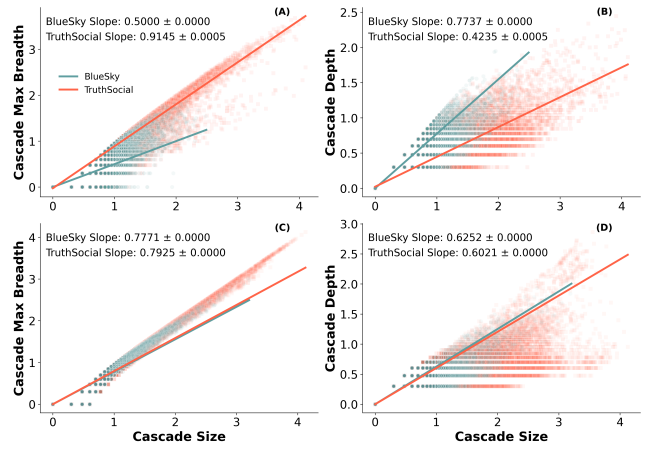


Figure 3: (A–B) Robust Linear Models (RLM) regressions of log-transformed cascade breadth (A) and depth (B) on log-transformed cascade size for reply cascades on BlueSky and TruthSocial. (C–D) Same regressions for repost cascades. Point clouds represent individual cascades (semi-transparent), and lines represent RLM fits.

Type	Platform	Cascades	Posts	Size 1	>50 posts
Reply	Bluesky	79,348	199,413	71.3%	0.3%
Reply	Truth Social	43,841	1,369,497	43.3%	9.0%
Repost	Bluesky	25,550	184,807	46.3%	2.3%
Repost	Truth Social	218,579	3,847,903	51.4%	5.0%

Table 1: Cascade counts and sizes. “Size 1” is the share of cascades consisting of a single post; the full size distribution is in Appendix Table A4.

This asymmetry suggests that while baseline amplification behaviors (reposts) are consistent, patterns of dialogic engagement diverge.

4.2 Mechanisms of Structural Divergence

The modeling results in Table 2 show clear differences in explanatory power across mechanisms. Model 2 (*Influencers*) indicates that cascades initiated by high-follower accounts on Truth Social are only marginally broader, and the platform–size slope gap remains largely unchanged relative to baseline. The broad–shallow structure of Truth Social cascades thus reflects widespread interaction patterns across the platform, not merely the behavior of a small elite.

Model 3 (*Ideological Structure*) is decomposed into three specifications. Composition alone (Model 3a) reduces divergence but leaves substantial residual differences (0.289 for breadth, -0.196 for depth). Alignment alone (Model 3b) likewise attenuates but does not eliminate platform differences (0.236 for breadth, -0.189 for depth). The combined model (3c) reduces both interaction terms by roughly five-sixths relative to the baseline, to 0.071 for breadth and -0.049 for depth against baseline values of 0.419 and -0.321 , showing that only when both ideological composition and cross-ideological alignment are considered together

Table 2: Interaction Effects of Size (Log) and Platform on Breadth (Log) and Depth (Log)

Model	Size (Log)	Size (Log) $\times$ TruthSocial	$pseudo R^2$	RMSE
Breadth (Log)				
Model 1 (Baseline)	0.5043 (0.0010)	+0.4187 (0.0012)	0.9176	0.1319
Model 2 (Influencers)	0.5042 (0.0010)	+0.4121 (0.0012) [†]	0.9182	0.1314
Model 3a (Composition)	0.3692 (0.0014)	+0.2887 (0.0019)	0.9364	0.1159
Model 3b (Alignment)	0.1713 (0.0012)	+0.2362 (0.0007)	0.9432	0.1095
Model 3c (Combined)	0.2137 (0.0016)	+0.0706 (0.0015)	0.9546	0.0979
Model 4 (Topic)	0.5140 (0.0014)	+0.4260 (0.0012)	0.9191	0.1307
Depth (Log)				
Model 1 (Baseline)	0.7320 (0.0008)	−0.3214 (0.0009)	0.8608	0.1031
Model 2 (Influencers)	0.7322 (0.0008)	−0.3134 (0.0009) [†]	0.8618	0.1027
Model 3a (Composition)	0.8432 (0.0011)	−0.1955 (0.0015)	0.8870	0.0929
Model 3b (Alignment)	0.9097 (0.0010)	−0.1889 (0.0006)	0.8926	0.0905
Model 3c (Combined)	0.9497 (0.0014)	−0.0491 (0.0013)	0.9085	0.0835
Model 4 (Topic)	0.7168 (0.0011)	−0.3302 (0.0009)	0.8623	0.1025

Notes: Robust linear models (Huber M -estimator) fitted with Huber’s proposal 2 scale estimate; asymptotic standard errors in parentheses. Every coefficient shown is more than 25 standard errors from zero, so significance stars are omitted as uninformative. $N = 123,189$ for every model.

[†] Model 2 replaces the binary platform indicator with a three-category variable (Bluesky, Truth Social non-influencer, Truth Social influencer); the entry shown is the Truth Social non-influencer contrast against Bluesky.

do platform-specific scaling gaps narrow this far. Model fit confirms this: pseudo- R^2 rises to 0.95 for breadth and 0.91 for depth in Model 3c, with corresponding reductions in RMSE. Because alignment is computed from the same reply edges that determine depth and breadth, we read this as statistical accounting rather than as a causal effect; the Limitations section returns to the point.

Model 4 (*Topic Variability*) shows no such effect. The platform-by-size interaction is essentially unchanged from the baseline once topic is included (0.426 against 0.419 for breadth, −0.330 against −0.321 for depth), indicating that differences in what the two platforms discuss do not account for the difference in how their cascades grow.

H1 receives limited support: influencer status accounts for only a small share of the platform difference, indicating that Truth Social’s broader and shallower reply cascades are not simply produced by high-follower users. H2 receives the strongest support: ideological composition and interaction alignment together account for most of the platform-specific divergence in scaling. H3 is not supported: adding topic leaves the platform-by-size interaction essentially unchanged.

4.3 Motif-Level Interaction Patterns

Chain motifs. We first examine chain motifs in reply cascades, which capture sequential exchanges where one reply receives another reply. Figure 4 shows how often three-node chain motifs occur relative to randomized reply-tree baselines. We organize these motifs into four ideological configurations: *fully aligned* chains, in which all three users share the same ideological label; *mid-shift* chains, in which the middle user differs ideologically from the first and final users; *end-shift* chains, in which the final user differs from the first two users; and *all-different* chains, in which all three users have different ideological labels.

Fully aligned Center-labeled chains are overrepresented relative to the null model on both platforms. Fully aligned Right chains fall within $|Z| < 3$ on both platforms: they are common in absolute terms on Truth Social, but no more so than the ideological composition of the platform already implies. Left-leaning chains diverge across platforms: Bluesky shows overrepresentation, whereas Truth Social shows underrepresentation.

The strongest contrasts appear in mid-shift “bounce-back” motifs. On Bluesky, all bounce-back chain motifs are overrepresented, so chains in which the middle user’s label differs from the two ends occur more often than the null model predicts, across every ideological pairing. On Truth Social, only Left↔Right bounce-backs are overrepresented, but at much higher magnitudes than on Bluesky. On Bluesky, then, back-and-forth across ideological lines occurs for every pairing of labels, whereas on Truth Social it is confined almost entirely to exchanges between left- and right-leaning users.

End-shift transitions (e.g., C–C–R, L–L–C) do not follow a uniform pattern. On Bluesky, end-shifts are generally at or below the rewired baseline, with L↔C links most underrepresented. On Truth Social, three end-shift motifs are overrepresented (R–C–C, R–R–L, L–C–C); each brings a center- or left-labeled user, the platform’s minorities, into a chain, for example a left-leaning reply at the end of a right–right exchange.

Star motifs. We next examine star motifs, which capture local branching in reply cascades: multiple users replying to the same parent post. Whereas chain motifs describe sequential extension across turns, star motifs describe how attention concentrates around a shared conversational object. Figure 5 summarizes standardized motif scores for three-node star motifs in reply cascades. Right-aligned (R–R–R)

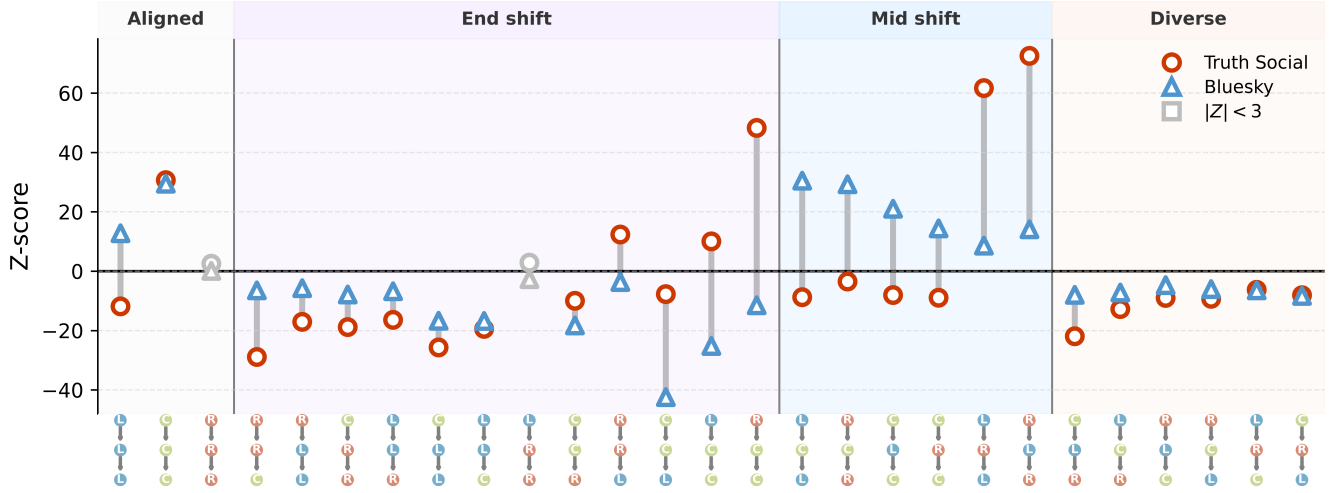


Figure 4: **Chain motif analysis of cross-platform reply structures.** Standardized differences (Z-scores) in the frequency of three-node chain motifs between observed cascades and a randomized null model. Node colors mark ideology (Left = blue, Center = green, Right = red); background shading groups motifs as aligned, mid-shift (middle node differs), end-shift (first or last node differs), or all-different. Circles represent Truth Social, triangles Bluesky; gray symbols denote values with $|Z| < 3$.

and center-aligned (C-C-C) stars are strongly overrepresented on Truth Social but suppressed on Bluesky, while left-aligned (L-L-L) stars show $|Z| < 3$ on either platform. This pattern suggests that homophilic clustering is not simply a majority-group phenomenon. Despite being the majority on Bluesky, left-leaning users do not cluster strongly in star formation. Truth Social also shows overrepresentation of several mixed configurations (L-R-R, L-C-C, C-R-R, R-L-L). **Most place a numerically rare ideology at the root and a more common pair among the repliers, which on Truth Social means left-labeled roots with right-labeled repliers.**

Summary. The motif analysis shows that cross-ideological interaction is not a single structural process. On Truth Social, disagreement more often takes a branching form: multiple users respond to the same parent post, producing local breadth without sustained sequential exchange. On Bluesky, disagreement more often takes a chaining form: replies build on one another, producing depth through multi-turn interaction. Thus, the key difference is not whether ideological difference appears in conversations, but what shape it takes when it appears.

5 Discussion

This study compares how candidate-centered conversations about the 2024 U.S. presidential election unfold on two ideologically distinct but technically comparable platforms, Truth Social and Bluesky. Whereas most prior work emphasizes how far or how fast information spreads, we focus on structural form: whether interactions broaden outward through shallow branching or deepen through sequential layering. By analyzing both reposts and replies, we show that reposts scale similarly across platforms, but reply cascades diverge sharply: Truth Social supports broader, shall-

lower exchanges, while Bluesky produces narrower, deeper threads. **These results show that platforms with similar reply and repost affordances can exhibit different conversational dynamics, depending on the social and ideological composition of the communities that use them.**

Our modeling results show that ideological composition and alignment together **statistically account for** most of the platform divergence: **cross-ideological exchanges are associated with greater depth, and more homogeneous cascades with flatter structures.** Yet alignment alone is insufficient. Even when disagreement occurs, reply structures on Truth Social often remain shallow. Motif analysis clarifies this pattern: on Truth Social, disagreement frequently appears as star motifs—parallel replies directed at a single post. On Bluesky, disagreement more often takes the form of chain motifs, where replies build sequentially and support multi-turn interaction.

Motif results highlight the combined role of interactional structure and user composition. On Truth Social, **left-leaning users are a small minority of the users we observe**, yet appear disproportionately in Left↔Right bounce-back motifs: these are strongly overrepresented relative to the null model while remaining rare in absolute terms. **This is why the model needs composition as well as alignment: how often exchanges cross ideological lines also depends on how many users of each ideology take part, which is what composition measures.**

Center-labeled users appear disproportionately in different local configurations across the two platforms. On Bluesky they occur more often in bounce-back motifs, as the targets of cross-ideological replies; on Truth Social they appear more often in several end-shift configurations.

A further implication is that diffusion differs not only in reach, speed, or volume, but also in the structure of partic-

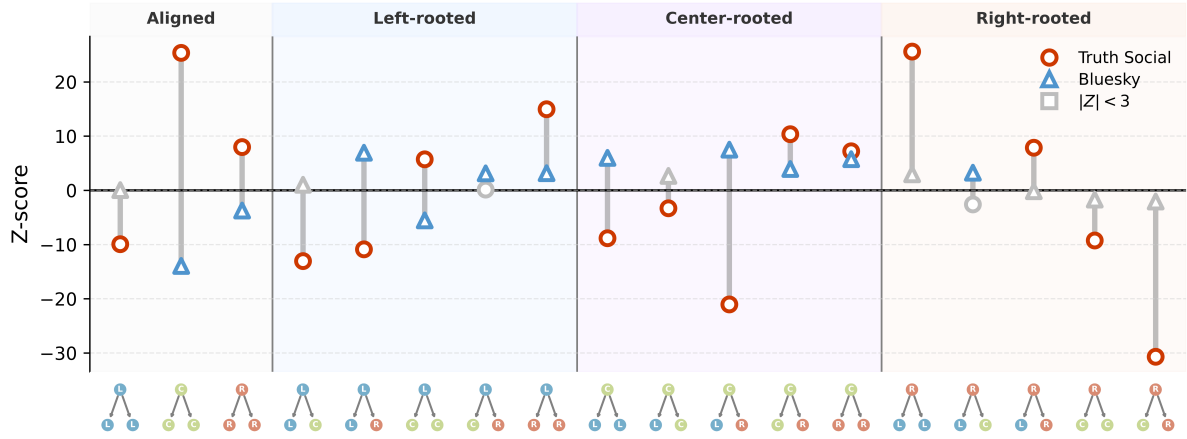


Figure 5: **Star motif analysis.** Standardized Z-scores of three-node star motif frequencies relative to degree-preserving null models. Circles mark Truth Social, triangles Bluesky; gray symbols indicate values with $|Z| < 3$.

ipation. A cascade can become large through broad, shallow attention or through deeper sequential exchange, and these forms imply different kinds of political engagement. This distinction is especially important in a fragmented platform ecosystem, where newer platforms may share familiar Twitter-like affordances while supporting different conversational dynamics. By comparing reposts and replies separately, our study shows that amplification can scale similarly across platforms even when conversation diverges. Political diffusion should therefore be understood not only as whether information spreads, but as how participation is organized once it does.

The two cascade shapes represent different forms of conversational organization. Wide, shallow reply cascades consist of many users responding to the same post with little subsequent exchange among them; narrower, deeper cascades involve replies that themselves receive replies across multiple turns. Neither is inherently more desirable, but they distribute attention differently, and the difference arises here between platforms with similar reply mechanics.

Methodologically, comparisons based on aggregate activity or cascade size alone would record Truth Social as simply more active and would miss this divergence entirely, because it appears only after conditioning on size. Structural measures of the kind used here are available wherever thread structure is recorded, and separating the diffusion patterns that recur across platforms from those that track the composition of a particular community is what makes findings portable to the next platform.

In our data, repost cascades scale similarly across platforms, whereas differences in reply structure are closely associated with ideological composition and interaction alignment.

6 Limitations

The alignment ratio is computed from the same reply edges that determine cascade depth and breadth. Cross-ideological

exchanges may contribute to longer sequential discussions, but sustained conversations may also create conditions under which additional cross-ideological exchanges occur. We therefore interpret ideological alignment as associated with cascade structure rather than as a causal determinant of it.

Three further constraints bear on interpretation. Our measure of ideological alignment is coarse, relying on a simplified typology that may not capture subtler dimensions of identity. Relatedly, our user-level ideology labels should be interpreted as coarse positions within the sampled Biden and Trump discourse rather than as context-independent measures of users’ full political ideology. Because the platforms also differ in user base and possibly in unobserved moderation, we treat the differences as emergent patterns in public interaction traces rather than as effects of platform design. And we use “technical similarity” in a deliberately narrow sense: both platforms provide Twitter-like affordances for posting, reposting, and threaded replying, which allows us to construct comparable cascade objects, though they differ in feed ranking, moderation, and recommendation architecture (see Appendix M).

These considerations limit the scope of generalization. Our sampling frame is candidate-centered and restricted to one month of the 2024 U.S. presidential campaign. The ideological composition observed here therefore reflects both the platform context and the particular political conversations included in the sample, and should not be interpreted as representative of either platform’s full user base. We also cannot determine whether the same structural patterns extend to other political topics or to non-political discussion. Our results therefore support a narrower conclusion: in conversations about U.S. politics, platforms with similar reply mechanics can exhibit different conversational structures conditional on cascade size, and these differences are associated with the ideological composition and interaction patterns observed within those conversations. Assessing how broadly these patterns generalize will require comparable analyses across platforms, countries, time periods, and top-

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 - (b) Do your main claims in the abstract and introduction accurately reflect the paper's contributions and scope?
Yes.
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Yes.
 - (d) Do you clarify what are possible artifacts in the data used, given population-specific distributions?
Yes.
 - (e) Did you describe the limitations of your work?
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- (b) Did you describe any potential participant risks, with mentions of Institutional Review Board (IRB) approvals?
Not applicable.
- (c) Did you include the estimated hourly wage paid to participants and the total amount spent on participant compensation?
Not applicable.
- (d) Did you discuss how data is stored, shared, and deidentified?
Yes.

A Power-Law Analysis of User-Level Activity Distributions

We analyzed the distribution of user-level activity—specifically, the number of followers, reposts, and replies per user—across both platforms using power-law fits. Figure A1 shows the complementary cumulative distribution functions (CCDFs) of these metrics in log-log scale, overlaid with fitted power-law tails (dashed orange lines). For each distribution, we report the estimated power-law exponent γ and the lower cutoff $x_{\min}$ above which the power-law holds.

The distribution of follower counts reveals strong skewness on both platforms. On Truth Social, a small number of users amass an exceptionally large following, with $\gamma = 1.76$ and $x_{\min} \approx 13,515$. Bluesky shows a slightly less extreme concentration ($\gamma = 1.84$, $x_{\min} \approx 1,498$). Repost and reply behaviors also exhibit heavy-tailed patterns. On Bluesky, replies decay rapidly ($\gamma = 3.02$) while reposts follow a shallower slope ($\gamma = 2.43$). Truth Social displays flatter distributions for both reposts ($\gamma = 1.84$) and replies ($\gamma = 1.79$), consistent with broader participation from users in redistributing and responding to content.

Exclusion of Highly Followed Users for Robustness.

To identify unusually influential users, we examined the follower-count distribution of every account in our data (134,100 on Truth Social; 41,566 on Bluesky). Both distributions are heavy tailed, but their upper tails differ in kind. On Truth Social the top eight accounts are separated from the body of the distribution: the largest (Donald Trump, 7,107,831 followers) has 56 times the followers of the platform’s 99.9th-percentile account (127,082), the eighth-largest still holds 2,170,535 followers, and together these eight hold 16.4% of the total follower count. On Bluesky no comparable class exists: the largest account (1,230,128) sits

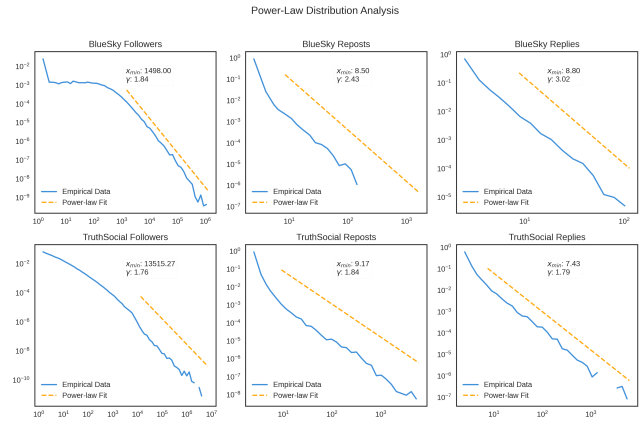


Figure A1: Follower count distributions across platforms.

five times above the 99.9th percentile (238,048), continuous with the rest of the distribution, and the top eight hold 6.1% of the total. The eighth-largest Truth Social account exceeds the largest Bluesky account.

We therefore designate the eight separated Truth Social accounts as influencers and apply no exclusion on Bluesky. A symmetric rule, such as excluding the top n percent on both platforms, would not equalize the comparison: on Bluesky it would remove accounts continuous with the rest of the distribution, while on Truth Social any n large enough to capture the separated class would also sweep in ordinary accounts.

B Ideology Estimation: Validation and Robustness

Ideological Distribution of Root Post Authors.

Figure A2 shows the ideological distribution of users who authored root posts. Bluesky exhibited a predominantly left-leaning user base (over 46,000 root posts from left-leaning users vs. 27,495 from right-leaning), while Truth Social was overwhelmingly right-leaning (37,639 right-authored root posts vs. 5,149 from left-leaning users and 1,053 from centrists).

Robustness to Ideology Thresholds.

We re-estimated Model 3 using alternative thresholds of 0.5 and 0.7 for classifying user ideology. Figure A3 shows how the share of users in each category changes with the threshold: center-labeled users remain at every threshold, including 0.5, and grow as the threshold rises. Table A1 reports the fit and the platform-by-size interaction at each threshold. Against the baseline model without ideology measures, whose platform interaction on this sample is 0.417 for breadth and -0.321 for depth, composition and alignment account for 63% and 67% of the platform difference at a threshold of 0.5, 84% and 86% at 0.6, and 98% and 96% at 0.7. The share rises with the threshold because a stricter threshold sorts more users into the center category, which composition then separates from the partisan

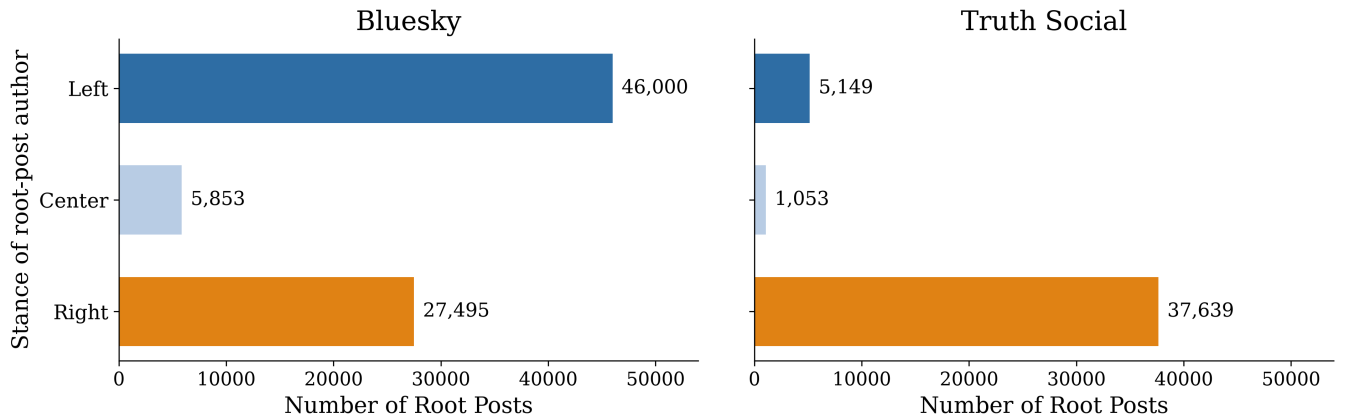


Figure A2: Distribution of root post partisanship across platforms. Bar plots show the number of root posts authored by users with left-leaning, centrist, or right-leaning stance labels. On *Bluesky*, left-leaning users dominate root-post activity. *Truth Social* exhibits a pronounced right-leaning skew. These distributions reflect the ideological clustering of user bases on each platform.

groups. Pseudo- R^2 values are uniformly high and error metrics change little. We also applied an unsupervised natural breakpoint detection algorithm to the ideology distribution, which identified 0.67 as a statistically distinct split point; we adopt 0.6 as our default to avoid excessive sparsity.

Class-Specific Validation Uncertainty.

All agreement figures use the same 200 validation replies. The two annotators agree at Cohen’s $\kappa = 0.77$ (bootstrap 95% interval $[0.70, 0.85]$), and the model agrees with them at $\kappa = 0.61$ (Annotator 1) and 0.74 (Annotator 2). For precision and recall we treat the human label as ground truth, using the 171 items on which both annotators agree; Table A2 reports these by class and platform with bootstrap intervals, and overall accuracy against this consensus label is 0.86.

The errors are not uniform across classes, and the pattern matters for interpreting our composition estimates: the model over-assigns each platform’s ideological minority. Precision is 0.64 for right-leaning labels on Bluesky and 0.64 for left-leaning labels on Truth Social, against 0.98 and 0.93 for the largest class on each platform. Because the minority-class estimates rest on relatively few consensus cases, we distinguish between the direction of the observed bias and its precise magnitude: the two minority cells rest on 8 and 17 consensus items and their precision intervals are correspondingly wide (for right on Bluesky, $[0.33, 0.91]$). Bootstrapping the difference between minority-class and majority-class precision within each platform gives -0.34 on Bluesky (one-sided $p = 0.017$) and -0.29 on Truth Social ($p = 0.004$), so the direction is supported on both platforms. Treating each annotator’s labels alone as ground truth, instead of the consensus, gives lower precision throughout, but the minority labels remain the least precise on both platforms. Uneven performance across the ideological spectrum is a documented property of language-model annotators (Vallejo Vera and Driggers 2025).

The misclassified items point to two sources of error. First, the model sometimes treats a post that describes the

right as a right-leaning stance, and a post that describes the left as a left-leaning one. What a post discusses and where its author stands are related in complex ways: a description of the right can come from either side, and so can criticism of it, since a reply complaining that Republican leaders are too soft on immigration criticizes the right from the right. Second, some replies carry little substantive content. A reply such as “What do we do?” can be read as supportive or as critical of the post it answers, depending on who writes it, which leaves room for error even with the preceding posts as context.

Robustness to Ideology Label Noise.

Because the stance labels contain measurement error, we tested whether the central result survives label noise at the measured rates. Misclassification by automated classifiers biases downstream regression even when overall accuracy is high (TeBlunthuis, Hase, and Chan 2024), and the same holds for language-model labels used as surrogates (Egami et al. 2023). Those authors correct the bias analytically with a gold-standard subsample; we instead propagate the measured error through the full pipeline, because our labels enter through composition, alignment, splines, and interactions rather than as a single regressor. For each of 100 draws per scenario, every user’s label was replaced by a draw from the measured confusion distribution, cascade-level composition and alignment were recomputed, and Model 3c was refit. We use the confusion matrix because it measures the model’s errors against human labels; the model’s own confidence in each post has not been validated against human judgment.

We ran this under four error processes: the error rates measured on each platform; one set of error rates pooled across both platforms, which applies to each platform’s dominant class an error rate measured mostly on the other platform; error rates re-estimated from a resampled validation sample in each draw, which carries the uncertainty of the small validation sample through; and the error rates obtained when a single annotator’s labels, instead of the con-

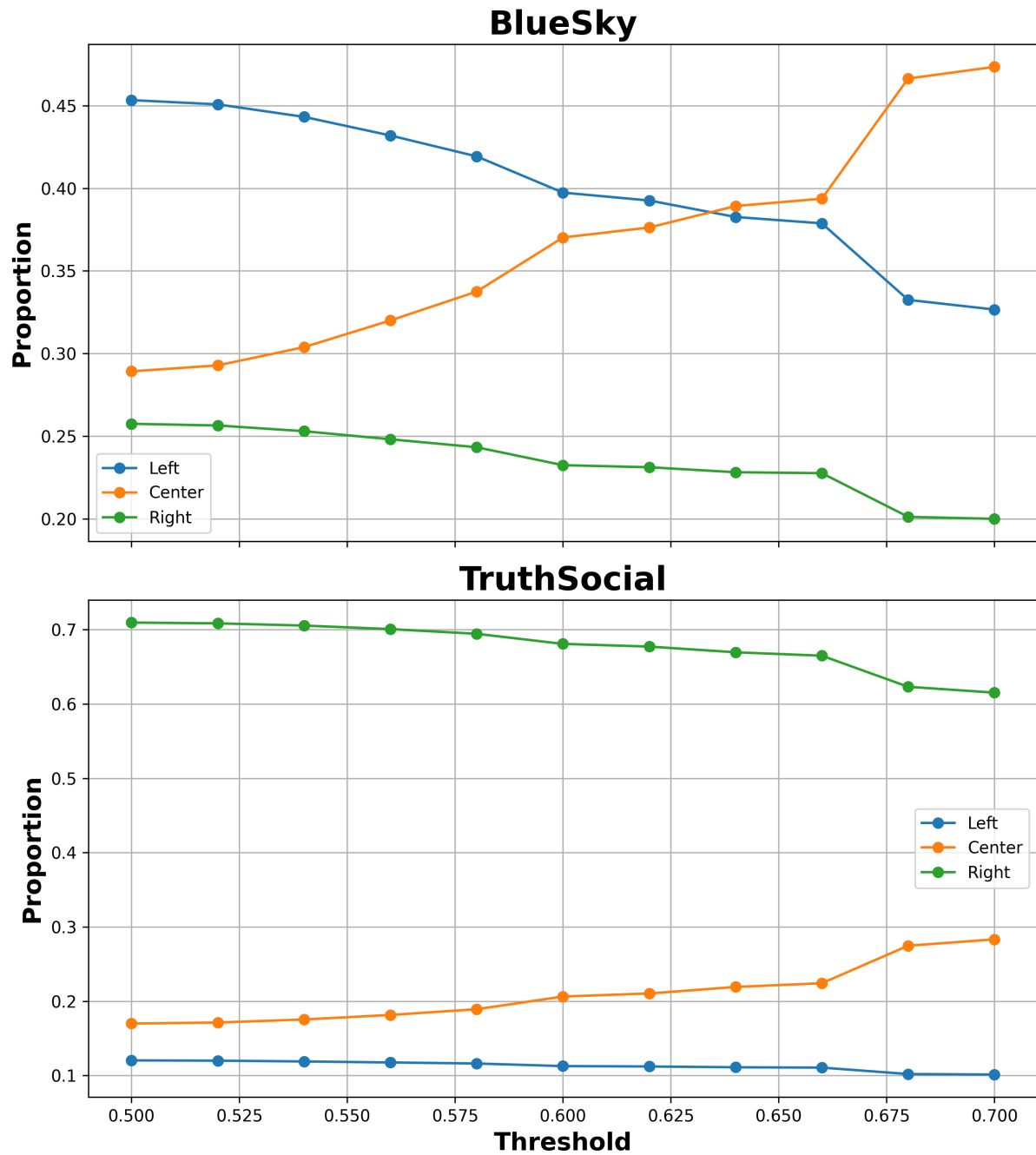


Figure A3: User ideology proportions and consistency across thresholds on *Bluesky* and *Truth Social*. Left-leaning users dominate at lower thresholds on *Bluesky*; right-leaning users represent the majority on *Truth Social* at all thresholds. Center-labeled users increase at higher thresholds on both platforms.

Table A1: Robustness check using alternative thresholds for defining user-level ideology. Results show that the main findings are stable across thresholds (0.5, 0.6, 0.7): platform gaps in cascade breadth and depth are consistently reduced once ideological structure is accounted for. Entries are Model 3c (composition and alignment jointly) fitted with Huber’s proposal 2 scale, matching Table 2; $n = 120,693$ cascades for which reply-tree ideology shares could be recomputed at each threshold.

Threshold	Breadth (Log)				Depth (Log)			
	Pseudo- R^2	MAE	RMSE	Size $\times$ TS	Pseudo- R^2	MAE	RMSE	Size $\times$ TS
0.5	0.954	0.044	0.100	0.155	0.908	0.039	0.084	-0.107
0.6	0.955	0.047	0.098	0.066	0.908	0.040	0.084	-0.045
0.7	0.957	0.045	0.096	-0.008	0.913	0.038	0.082	0.013

Scope	Class	Precision [95% CI]	Recall	n
Pooled	left	0.68 [0.54, 0.82]	0.80	35
Pooled	center	0.96 [0.90, 1.00]	0.85	59
Pooled	right	0.89 [0.81, 0.95]	0.90	77
Bluesky	left	0.74 [0.53, 0.93]	0.78	18
Bluesky	center	0.98 [0.92, 1.00]	0.89	45
Bluesky	right	0.64 [0.33, 0.91]	0.88	8
Truth Social	left	0.64 [0.42, 0.83]	0.82	17
Truth Social	center	0.91 [0.69, 1.00]	0.71	14
Truth Social	right	0.93 [0.86, 0.98]	0.90	69

Table A2: Class-specific and platform-specific model performance against human consensus labels (171 of 200 validation items on which both annotators agree). Intervals are percentile bootstrap 95% intervals over 2,000 resamples; n is the consensus support per class.

sensus, are treated as ground truth, under which precision for right-leaning labels on Bluesky falls to 0.33. The baseline model without ideology measures does not use the labels, so its platform difference (0.419 for breadth and -0.321 for depth) is the same in every draw; what label noise can change is how much of that difference composition and alignment account for. Table A3 reports this share. With the original labels it is 84% for breadth and 86% for depth. In the median draw of each process it stays between 70% and 86%, and in the least favorable of all 400 draws it is 66%. The pooled process lowers the share the most, and the per-platform and resampled processes, which are closest to the error we measured, leave it near its original value. Label noise therefore lowers the share under some assumptions, but composition and alignment account for most of the platform difference in every simulation. All fits use Huber’s proposal 2 scale, as in Table 2.

Ideological Composition Compared With Prior Estimates.

Our labels place 41.4% of Bluesky users in our sample on the left, 34.4% at center, and 24.2% on the right; on Truth Social the shares are 11.5% left, 19.3% center, and 69.2% right. The Bluesky figures show a larger right-leaning presence than platform-wide estimates: Quelle and Bovet (2025) classify 75.3% of Bluesky users as left of center and 4.8% as right of center, and Nogara et al. (2026), scoring active users by the leaning of the domains they share, report 74.61% left-

Error process	Breadth		Depth	
	Median	Worst	Median	Worst
Original labels	84%		86%	
Per-platform rates	86%	81%	84%	78%
Pooled rates	74%	71%	70%	66%
Resampled validation	86%	72%	84%	69%
Single annotator	79%	72%	76%	69%

Table A3: Share of the baseline platform difference that composition and alignment account for under simulated label noise, 100 draws per error process. “Worst” is the least favorable draw.

leaning, 18.17% centrist, and 7.21% right-leaning.

Two differences account for most of this gap. First, the samples differ: platform-wide estimates describe all users, most of whom post little political content, and are typically derived from shared news domains, whereas our sample is restricted to accounts posting in a conversation mentioning Biden or Trump during one campaign month, labeled from post text. Second, part of the gap is measurement error. Applying a prevalence correction that inverts the measured confusion matrix lowers the estimated right-leaning share on Bluesky from 24.2% to 17.9% and the left-leaning share on Truth Social from 11.5% to 1.9%. Both corrections move in the expected direction, but for Bluesky a substantial gap remains after correction, so label error alone does not account for it. The remaining discrepancy may partly reflect differences in sampling frame, since a candidate-centered conversation sample is not a sample of the platform population. These corrected values are indicative rather than precise, because the correction inherits the imprecision of the minority cells in Table A2.

C Descriptive Statistics

Table A4 gives the full size distribution behind the main-text summary in Table 1.

D Robustness of Repost Cascade Reconstruction

Any reconstruction rule could in principle shape the structural comparison (DeVerna et al. 2024), so we re-derived every repost cascade under alternative rules and re-estimated the platform-by-size interaction (β_3 in Eq. 1) for each. Following Bakshy et al. (2011), who report results under first,

Type	Platform	Cascades	Posts	Cascades by size, n (%)			
				1	2–5	6–50	>50
Reply	Bluesky	79,348	199,413	56,552 (71.3)	16,823 (21.2)	5,710 (7.2)	263 (0.3)
Reply	Truth Soc.	43,841	1,369,497	18,967 (43.3)	11,057 (25.2)	9,868 (22.5)	3,949 (9.0)
Repost	Bluesky	25,550	184,807	11,821 (46.3)	8,584 (33.6)	4,546 (17.8)	599 (2.3)
Repost	Truth Soc.	218,579	3,847,903	112,266 (51.4)	64,991 (29.7)	30,490 (13.9)	10,832 (5.0)

Table A4: Cascade counts and size distributions. For reply cascades, size is the number of posts in the thread including the root. For repost cascades, size is the number of reposts.

most recent, and split attribution, we compare several rules. Every rule we consider selects each repost’s parent from the same candidate set: the original post, plus earlier reposters whom the reposting user follows. We compare the rule used in the paper, which attaches a repost to the original post whenever the user follows its author, with rules that choose among the earlier reposters the user follows and use the original post only when there are none: the earliest, the most recent, or a uniformly random one (40 draws). As a check, each is also run with the repost list order reversed.

Figure A4 summarizes the result over 244,129 repost cascades (4.03 million reposts). Across all rule variants, counting each random draw separately, the breadth interaction lies between -0.012 and $+0.031$ and the depth interaction between -0.083 and $+0.028$. The same estimator applied to reply cascades gives $+0.170$ (breadth) and -0.182 (depth), exceeding the most extreme repost estimate under any rule by a factor of five and two respectively. The original reconstruction rule yields the largest depth difference, in absolute value, of any rule, and for breadth every rule gives a value within 0.031 of zero. The similarity in repost scaling is therefore robust to the reconstruction rules examined here. (Because log breadth and log depth of small trees take few distinct values, the scale estimate of a Huber robust regression degenerates on repost cascades; the interactions in this analysis are estimated by OLS with HC3 standard errors.)

The rule has limited room to matter in the first place: 55% of Bluesky reposts and 59% of Truth Social reposts have no eligible prior reposter and attach to the root under every rule, and a further 21% and 12% have exactly one candidate. A second concern is that the follower network is a snapshot without edge creation dates, so a follow created after a repost could produce a spurious parent. The follower lists were collected together with the posts and reposts, one day apart, on both platforms, so few follow edges should post-date the reposts they explain and the error applies to both platforms alike. As a check, under the most-recent-reposter rule we deleted 5% of follow edges at random, likely more than such timing error could produce; the breadth interaction moves from 0.031 to 0.028 and the depth interaction from 0.004 to 0.005, both far below the reply values.

E Two-User Exchanges in Reply Cascades

Depth would mean something different if it largely measured back-and-forth exchange between the same two accounts rather than extended multi-party discussion. We therefore counted, for every reply, whether its author also

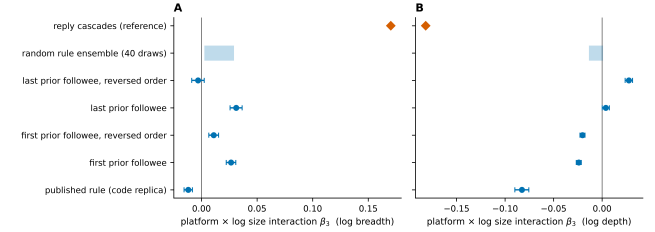


Figure A4: Platform-by-size interaction β_3 for repost cascades under alternative reconstruction rules (dots with 95% confidence intervals; shaded band, range over 40 random-rule draws), for log breadth (A) and log depth (B). The diamond marks the reply-cascade interaction under the same estimator.

wrote its grandparent post, that is, an $A \rightarrow B \rightarrow A$ pattern. Among all cascades, 9.3% on Bluesky and 20.8% on Truth Social contain at least one such exchange, reflecting mainly that Truth Social cascades are larger. Conditioning on cascades of depth two or more, the share is 68.8% on Bluesky and 70.0% on Truth Social, and such exchanges account for 20.0% and 15.6% of reply edges respectively. Dyadic exchanges are therefore common on both platforms to a similar degree. Whole threads between only two accounts differ more: among threads with a reply to a reply, 28.4% on Bluesky and 6.5% on Truth Social involve only two accounts, although Truth Social threads are much larger (a median of 18 participants, against 3). Refitting the models on threads with at least one reply after dropping every two-participant thread leaves most of the depth difference in the baseline model in place (-0.184 , against -0.222 among all threads with a reply), and composition and alignment still account for a majority of it (63% for depth and 59% for breadth). Two-person conversations therefore do not explain the depth difference.

F Motif Frequencies and the Null Model

Table A5 reports the raw observed frequency of every ideology-labeled three-node motif on each platform, alongside which the z-scores should be read. The two platforms differ by orders of magnitude in absolute motif volume, so a large z-score does not imply a common configuration.

The z-scores compare each observed count with its distribution over 100 randomized graphs per platform, built by double-edge swaps within each cascade: two reply edges are

Ideology triple	Star		Chain	
	Bluesky	Truth Soc.	Bluesky	Truth Soc.
L-L-L	13,436	1,423,162	6,519	4,747
L-L-C	20,482	1,361,062	2,659	3,270
L-L-R	3,869	4,721,187	346	11,883
L-C-L	18,970	1,695,796	4,966	3,727
L-C-C	31,468	2,929,973	3,113	7,194
L-C-R	6,179	6,432,294	290	8,869
L-R-L	3,751	4,837,988	471	26,476
L-R-C	5,846	5,384,674	191	10,673
L-R-R	1,462	20,194,975	88	40,264
C-L-L	16,997	158,383	1,533	2,040
C-L-C	36,051	137,961	6,197	3,390
C-L-R	8,058	437,959	249	5,929
C-C-L	30,243	142,810	2,930	3,421
C-C-C	74,502	197,087	25,971	13,261
C-C-R	15,559	458,039	980	8,179
C-R-L	6,786	466,118	161	7,371
C-R-C	15,173	458,308	1,518	8,093
C-R-R	4,093	1,929,621	171	16,393
R-L-L	407	11,197,554	89	15,565
R-L-C	841	6,961,901	111	14,127
R-L-R	281	24,241,501	307	66,065
R-C-L	778	8,183,288	111	15,549
R-C-C	1,735	6,435,618	384	30,188
R-C-R	625	19,234,403	819	48,366
R-R-L	238	23,352,704	74	45,060
R-R-C	448	15,651,387	163	30,149
R-R-R	224	64,855,781	357	128,058
Total	318,502	233,481,534	60,768	578,307

Table A5: Observed counts of ideology-labeled three-node motifs (L = left, C = center, R = right). For chains the triple is ordered along the reply path; for stars it lists the root followed by the two leaves.

chosen at random and the replies they lead to are exchanged, together with those replies’ ideology labels, with $5m$ swap attempts in a cascade of m edges. This holds fixed the set of cascades and their sizes, each post’s number of replies, and the ideology attached to every participant, while randomizing who replies to whom. Motifs of this kind are not mutually independent: a chain of four nodes contains two overlapping three-node chains. This dependence does not bias the z-scores, because the randomized graphs are counted with the identical procedure, so overlap affects observed and expected counts alike. It does affect how several motifs are read together: neighboring motif types share substructure, so their z-scores are correlated and we read them as a pattern across families rather than as independent evidence.

G Topic Model Details

The topic model was fit over 125,623 root posts pooled across platforms. HDBSCAN initially assigns low-density documents to an outlier class; we applied BERTopic’s outlier reduction step with default settings, which reassigns those documents to their nearest topic, leaving 21 documents (0.02%) unassigned. Clustering yields 12 topics plus a residual outlier class. Manual labeling then produced the 11 co-

herent and interpretable categories used in the analysis: the residual class, which retains only 21 posts, is excluded, and two near-duplicate pro-Trump categories are treated as one. Every cascade in the modeling sample carries one of these 11 labels. Because topic variation accounts for relatively little of the platform difference (Model 4), uncertainty introduced by outlier reassignment is unlikely to affect the main comparison.

H Modeling Strategy and Regression Diagnostics

Initial Ordinary Least Squares (OLS) regression revealed strong heteroskedasticity, with residuals showing a fan-shaped pattern as a function of fitted values (Figure A5). Residuals also exhibited heavy tails resembling a t-distribution. Although Cook’s Distance analysis identified a small number of observations with relatively higher influence, the vast majority fell well below the threshold of $4/n$. The heavy-tailed, heteroskedastic residuals motivate robust estimation, and all models reported in the main text are robust linear models with the Huber loss. Because 61.3% of reply cascades consist of a root post with no replies, and these sit exactly at the origin of the log-log scaling plot, the default median-absolute-deviation scale estimate collapses toward zero; standard errors computed from it are not interpretable. We therefore fit with Huber’s proposal 2 scale, which does not degenerate here, and report a sensitivity analysis across samples and scale estimators below. The one analysis estimated by OLS with HC3 errors is the repost reconstruction robustness check, where lattice-valued outcomes make the Huber scale estimate degenerate.

Sensitivity of the Modeling Specification.

Two features of the reply-cascade data warrant an explicit check. First, 61.3% of reply cascades consist of a root post with no replies (71.3% on Bluesky, 43.3% on Truth Social). These are structurally degenerate: breadth and depth are fixed by definition, so they sit exactly at the origin of the log-log scaling plot. Second, because this mass of exactly-fitted points makes the median absolute deviation of the residuals vanish, the default scale estimator collapses toward zero, and standard errors computed from it are not interpretable.

Table A6 therefore re-estimates the baseline model and Model 3c across the four combinations of sample and scale estimator. The baseline interaction is smaller when single-post cascades are excluded, as expected once the degenerate points are removed, but the substantive result is stable: including composition and alignment absorbs between 81% and 94% of the baseline platform difference in every specification, for both breadth and depth. Table 2 reports the Huber proposal 2 fit on all cascades.

I Spline Transformations

To examine potential nonlinear predictor effects, we inspected partial residual plots for the three ratio predictors (Figure A6). Each ratio enters the model both on its own and interacted with $\log(\text{size})$, so each panel plots the predictor’s full contribution against a straight line carrying the

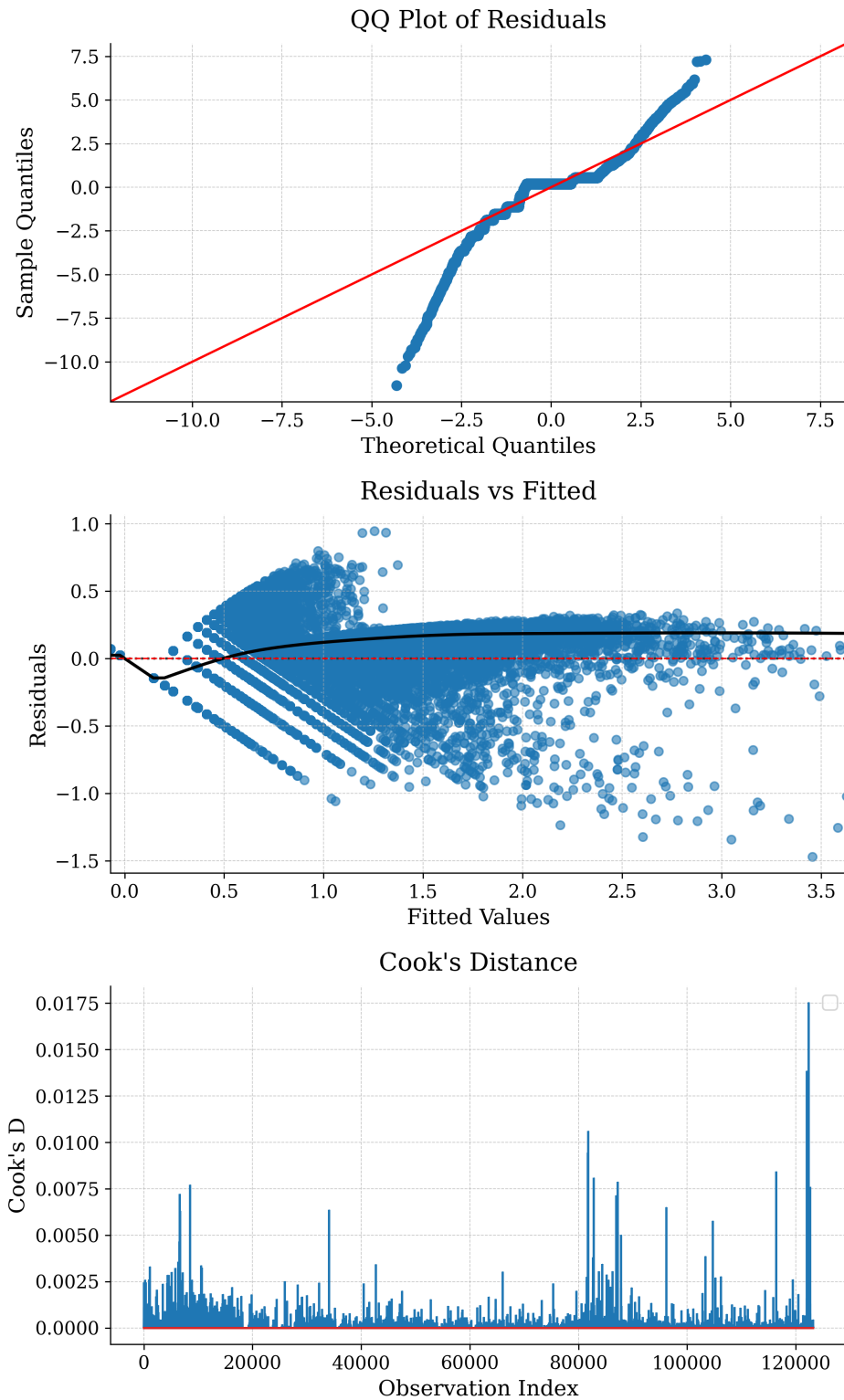


Figure A5: **Regression diagnostic plots.** Top: quantile–quantile (QQ) plot of residuals indicating heavy-tailed error distribution. Bottom: residuals vs. fitted values showing a fan-shaped pattern (heteroskedasticity). LOESS smoother overlaid.

Sample	Scale est.	Outcome	Baseline	Model 3c	Absorbed
All	MAD	breadth	0.398	0.023	94%
All	MAD	depth	-0.334	-0.055	84%
All	Huber	breadth	0.419	0.071	83%
All	Huber	depth	-0.321	-0.049	85%
Size > 1	MAD	breadth	0.187	0.035	81%
Size > 1	MAD	depth	-0.223	0.019	91%
Size > 1	Huber	breadth	0.181	0.015	92%
Size > 1	Huber	depth	-0.217	0.029	87%

Table A6: Platform-by-size interaction before and after adding ideological composition and alignment, across samples and scale estimators. “Absorbed” is the proportion of the baseline interaction removed by Model 3c. $n = 123,189$ for all cascades and 47,670 for cascades with more than one post.

slope that contribution has at mean cascade size. Departure of the smoother from that line is what a spline would capture.

The left-user ratio tracks its linear term closely across the range. The right-user ratio departs from it between roughly 0.6 and 0.95, and the alignment ratio departs from it across most of the range. The largest gap between smoother and line is 0.06 for the left-user ratio, 0.17 for the right-user ratio, and 0.23 for the alignment ratio.

We apply spline transformations (`bs` basis, `df=3`) to `author_left_ratio` and `author_alignment_ratio`; B-splines are preferred over natural cubic splines here because the data are confined to $[0,1]$ and used in interactions, where their local support provides greater numerical stability. This places a spline on the predictor with the least curvature and a linear term on one with more, so we report what the choice costs. Refitting Model 3c with all three ratios splined absorbs 98.8% of the baseline breadth interaction and 93.5% of the depth interaction; entering all three linearly absorbs 68.4% and 69.7%. The specification we report falls between these, at 83.1% and 84.7%. It is therefore the more conservative of the two spline choices, and the finding that ideological structure absorbs most of the platform gap holds wherever the splines are placed.

J Final Model Specification

Our final model specification includes: (1) a main effect of `log_size` and its interaction with `platform`; (2) `right_user_ratio` and spline-based transformations of `alignment_ratio` and `left_user_ratio`, interacted with both `log_size` and `platform`; and (3) topic category and outlier status as categorical moderators. All models were estimated using the `statsmodels` package in Python with robust standard errors.

K Robustness Checks: Alternative User Composition Specifications

To assess whether findings depend on how user composition is encoded, we compared four specifications: (1) left/right

shares only, (2) majority/minority shares only, (3) left/right with alignment, and (4) majority/minority with alignment. Table A7 reports results for both cascade breadth and depth.

Including alignment substantially improves fit for both outcomes (Breadth $R^2 \approx 0.95$; Depth $R^2 \approx 0.90$), confirming that interaction structure plays a central role in cascade geometry. The majority/minority framework gives slightly lower absolute error but slightly worse pseudo- R^2 and RMSE, so neither encoding dominates. Its residual $\log(\text{size}) \times \text{platform}$ interaction is smaller in the composition-only models and larger once alignment is included, which is expected: recoding composition by each platform’s dominant ideology absorbs part of the platform contrast directly into the composition terms, and that absorption is redundant once alignment is in the model. The attenuation conclusion is the same under both encodings. We retain the left/right specification as our main model because its coefficients carry a fixed ideological meaning across platforms.

L Marginal Effects of Ideological Structure

Figure A7 plots marginal effects of alignment ratio, left-user ratio, and right-user ratio on cascade structure. For **depth**, higher alignment is associated with shallower structures, consistent with the idea that ideologically homogeneous cascades sustain fewer extended exchanges. Cascades with fewer right-leaning participants exhibit greater layering, echoing the motif finding that right-leaning users rarely construct fully right-aligned chains. The marginal effects of left- and right-user ratios are asymmetric: increases in right-leaning participation are linked to sharper declines in depth than comparable increases in left-leaning participation. For **breadth**, higher alignment corresponds to wider cascades, indicating that homogeneous groups are more likely to generate simultaneous uptake at the same level. The left-user ratio exhibits an inverted-U relationship with breadth, while the right-user ratio shows a more linear pattern: as right-leaning users become increasingly dominant, cascades grow consistently broader.

M Platform Affordances and Technical Comparability

Our comparison rests on a narrow notion of technical comparability: we do not assume the platforms are identical exposure environments, but that they share Twitter-like affordances (posting, reposting, threaded replies) that enable construction of comparable cascade objects. Ranking systems, moderation decisions, and feed settings may affect which posts users see and which cascades grow; these are partly opaque and cannot be fully identified from public traces. Our analysis should be interpreted as an observational comparison of public interaction traces, not a causal estimate of platform design effects.

The technical lineage of both platforms provides additional justification. Bluesky is built on the AT Protocol, which also supports custom feed generation through a feed-generator architecture. Truth Social has been widely documented as Mastodon-derived, using open-source social net-

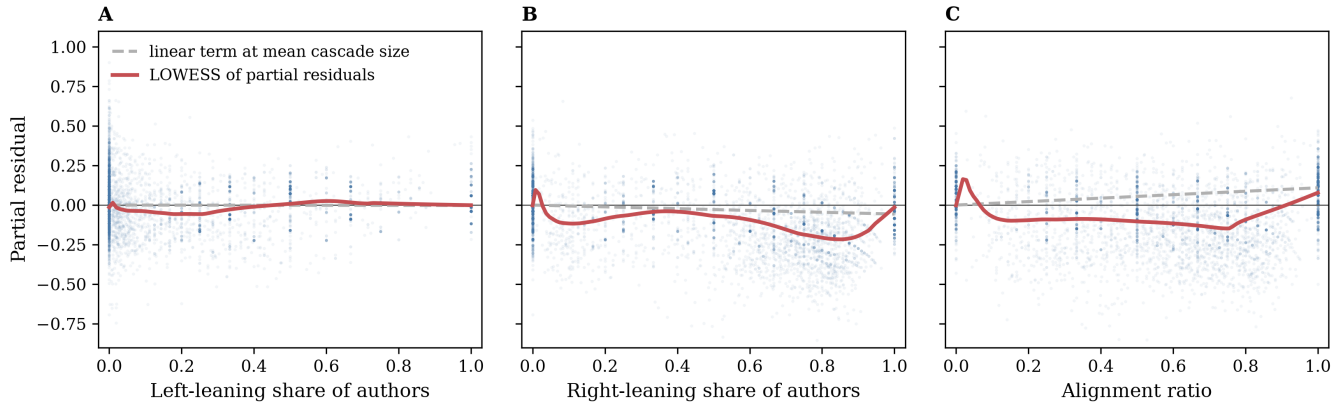


Figure A6: Partial residual plots under a linear baseline for (A) `author_left_ratio`, (B) `author_right_ratio`, and (C) `author_alignment_ratio`. Points are a random 20,000 of the 123,189 cascades; the fit uses all of them. The dashed line is the predictor's linear contribution evaluated at mean cascade size, the solid curve a LOWESS smoother of the partial residuals. Separation between the two is the nonlinearity a spline basis would absorb.

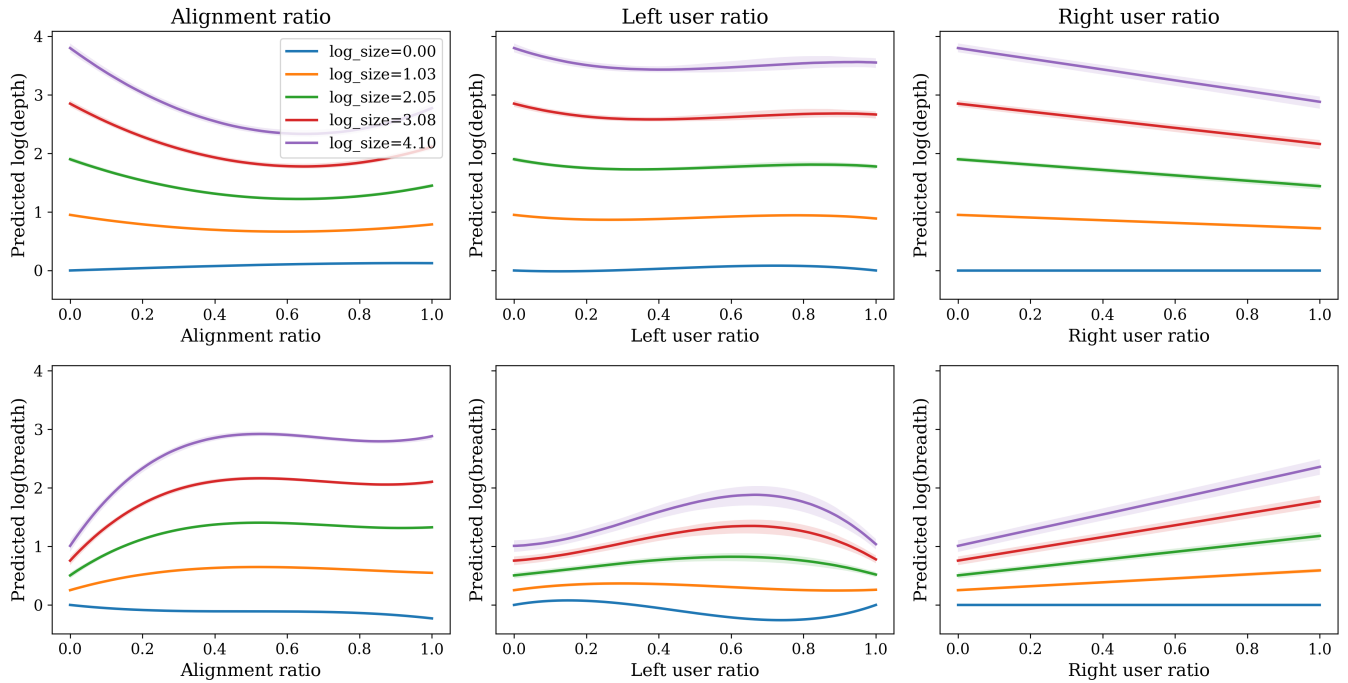


Figure A7: Marginal effects of ideological alignment, left-user ratio, and right-user ratio on cascade structure. Solid lines denote predicted `log(depth)` (top row) and `log(breadth)` (bottom row) at five equally spaced values of `log(size)`. Shaded bands represent 95% bootstrap confidence intervals. Homogeneous cascades (high alignment) tend to flatten into wide, shallow structures; heterogeneous cascades deepen through sequential exchanges.

Table A7: Comparison of user-composition encodings (Left/Right vs. Majority/Minority), with and without alignment. Robust Linear Models.

	Panel A: Composition only				Panel B: + Alignment (df=3)			
	Breadth (log_breadth)		Depth (log_depth)		Breadth (log_breadth)		Depth (log_depth)	
	L/R	Maj/Min	L/R	Maj/Min	L/R+Align	Maj/Min+Align	L/R+Align	Maj/Min+Align
Observations	120,693	120,693	120,693	120,693	120,693	120,693	120,693	120,693
Pseudo R^2	0.9367	0.9359	0.8861	0.8825	0.9548	0.9526	0.9078	0.9064
MAE	0.0645	0.0638	0.0500	0.0480	0.0468	0.0437	0.0402	0.0380
RMSE	0.1164	0.1172	0.0935	0.0949	0.0984	0.1008	0.0841	0.0848
$\beta_{\log(\text{size})}$	0.3718	0.3552	0.8411	0.8285	0.2134	0.4106	0.9478	0.8093
$\beta_{\log(\text{size}) \times \text{platform}(\text{ts})}$	0.2847	0.1250	-0.1913	-0.1093	0.0662	0.1216	-0.0455	-0.0970

Notes: L/R models the *left-user share* with a spline (df=3) and the *right-user share* linearly, interacted with log(size) and platform. Maj/Min re-codes composition by platform-dominant ideology (majority) vs. opposing (minority); majority share modeled with a spline (df=3), minority share linearly. Panel B additionally includes the *alignment ratio* (spline, df=3) and its interactions. **All eight models are fitted with the Huber loss and Huber’s proposal 2 scale, matching Table 2. The sample is the 120,693 cascades for which reply-tree ideology shares could be recomputed, so β_3 differs slightly in level from Table 2. Bold marks the better of the two encodings within each column pair. Coefficients are reported without standard errors because only the comparison between encodings is at issue and both share the estimator.**

working software with visual modifications. These lineages support structural comparability of cascade objects while reinforcing that ranking, moderation, and exposure should be treated as platform-specific and only partially observed.

memory each). Model inference was deployed using **vLLM** with **bfloat16 (BF16) precision**, enabling efficient batched inference while maintaining numerical stability for long-context inputs.

Table A8: Platform affordances and technical lineage relevant to cascade construction.

Dimension	Truth Social	Bluesky
Technical lineage	Mastodon-derived platform	Built on the open-source AT Protocol ecosystem
Posting affordance	Supports user-authored posts	Supports user-authored posts
Reposting affordance	Supports repost-style resharing	Supports repost-style resharing
Reply affordance	Supports threaded replies	Supports threaded replies
Cascade observability	Public repost and reply traces observable	Public repost and reply traces observable
Feed and ranking	Platform-specific visibility and ranking; not directly observed	Multiple feeds via feed-generator architecture; visibility varies
Moderation	Platform-specific moderation may affect observed patterns	Decentralized moderation and curation may affect patterns
Analytic implication	Enables cascade construction; exposure process not fully identified	Enables cascade construction; exposure process not fully identified

N Computing Environment

All experiments were conducted on a single-node, multi-GPU system equipped with **8 NVIDIA H100 GPUs (80GB**